

Tasks

1. Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

Step 1 :- Open Hosts File

C:\Windows\System32\drivers\etc\hosts

Step 2 :- Open notepad ---- run as a administrator ---- file save on above path -
----- and select type all files

Step 3 :- Add wrong Entry

10.0.0.123 www.spotify.com

Step 4 :- Save the file

Step 5 :- win + R ----- cmd ----- ipconfig /flushdns

Step 6 :-

```
C:\Users\cdiks>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\cdiks>ping www.spotify.com

Pinging www.spotify.com [10.0.0.123] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

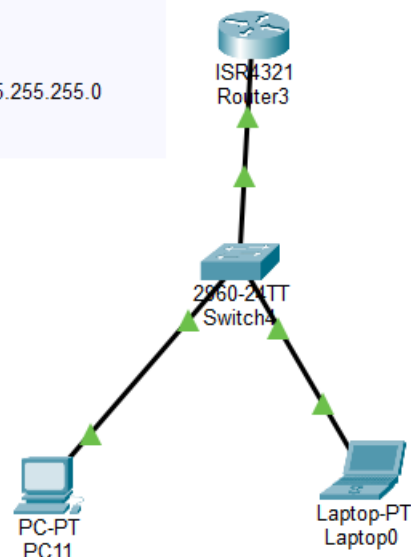
Ping statistics for 10.0.0.123:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

2. Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

Step 1 :- Configure Router Interface

```
en
conf t
interface gig0/0/0
ip address 10.0.0.1 255.255.255.0
no shutdown
exit

interface gig0/0/1
ip address 20.0.0.1 255.255.255.0
no shutdown
exit
```



Step 2 :- Configure DHCP Server with Wrong Gateway

- ip dhcp excluded-address 10.0.0.1 10.0.0.10
- ip dhcp excluded-address 20.0.0.1 20.0.0.10
- ip dhcp pool LAN1
- network 10.0.0.0 255.255.255.0
- default-router 10.0.0.254
- dns-server 8.8.8.8
- exit

- ip dhcp pool LAN2
- network 20.0.0.0 255.255.255.0
- default-router 20.0.0.254
- dns-server 8.8.8.8
- exit

Step 3 :- Automatic Ip (DHCP)

Step 4 :- pc cmd ----- ipconfig /all

Step 5 :- Fix the DHCP Configuration

- ip dhcp pool LAN1
- network 10.0.0.0 255.255.255.0
- default-router 10.0.0.1
- dns-server 8.8.8.8
- exit

- ip dhcp pool LAN2
- network 20.0.0.0 255.255.255.0
- default-router 20.0.0.1
- dns-server 8.8.8.8
- exit

Step 6: ping 10.0.0.1

3. Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Answer :- Computers on the network could open websites using IP addresses, but they could not open websites using names such as www.google.com. This showed that there was a DNS problem.

Step 1: Checked network connectivity using an IP address.

- ping 8.8.8.8

The ping was successful.

Step 2: Checked connectivity using a website name.

- ping www.google.com

The ping was successful.

Step 3: Verified DNS resolution using nslookup.

- nslookup www.google.com

The command successfully returned the IP address of the website.

Step 4: Checked DNS configuration.

- ipconfig /all

The DNS server settings were correct.

Conclusion:

The network and DNS services were working properly. Devices were able to access websites using both IP addresses and domain names. No DNS issue was found.

4. A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Answer :-

Problem:

The client device received an **APIPA address (169.254.x.x)** instead of a DHCP IP address. This means the device could not contact the DHCP server.

Possible Causes:

1. DHCP server is off.

2. Network cable is disconnected.
3. Switch port is down.
4. DHCP pool has no available IP addresses.
5. Wrong VLAN configuration.
6. Missing DHCP relay (ip helper-address).

Troubleshooting Steps:

1. Check the client's IP address using `ipconfig /all`.
2. Verify that the network cable is connected properly.
3. Check that the switch port and router interface are up.
4. Verify that the DHCP server is running.
5. Check the DHCP pool configuration and available IP addresses.
6. Verify VLAN settings and DHCP relay configuration.
7. Renew the IP address using:
 1. `ipconfig /release`
 2. `ipconfig /renew`
8. Test connectivity by pinging the default gateway.

Solution:

The DHCP configuration was corrected, and the client renewed its IP address. The device successfully received a valid IP address from the DHCP server and network connectivity was restored.

Result:

The APIPA address (169.254.x.x) was replaced with a valid DHCP-assigned IP address (for example, 192.168.1.10), and the issue was resolved.

5. Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Common DNS Issues

Issue	Symptoms	Cause	Solution
Wrong DNS Server	Website opens with IP but not name	Incorrect DNS setting	Set correct DNS server
DNS Cache Problem	Some websites don't open	Old/corrupt DNS records	Run ipconfig /flushdns
DNS Server Down	No website opens by name	DNS server unavailable	Check DNS server and network

Common DHCP Issues

Issue	Symptoms	Cause	Solution
APIPA Address (169.254.x.x)	No network access	DHCP server not responding	Check DHCP server and renew IP
Wrong Gateway	Internet not working	Incorrect gateway in DHCP	Configure correct gateway
No IP Available	New device gets no IP	DHCP pool is full	Increase IP range or free unused IPs

Useful Commands

Command	Use
ipconfig /all	View network settings
ipconfig /renew	Get a new IP address
ipconfig /flushdns	Clear DNS cache
nslookup google.com	Check DNS resolution
ping 8.8.8.8	Test network connectivity